

Mining the seas for lithium

Currently, 75 percent of the lithium in the world comes from an arid region in the Atacama desert known as the lithium triangle, stretching across Argentina, Bolivia and Chile. If the energy requirements of the humans have to be met by greener alternatives to fossil fuels, humanity will require lots of lithium in the future for electric vehicles which includes everything from mopeds, to tractors, to cars, to drone ships to cargo dirigibles. This demand simply cannot be met from the lithium triangle alone, which has geologists on the hunt for alternative sources, going as far as conducting underwater surveys prospecting for mineral deposits on the ocean floor. A team of



researchers from The University of Chicago have come up with a novel way of farming the lithium from the

waters of the oceans itself. The process is called electrochemical intercalation, that attracts lithium ions to an array of electrodes using an electric field. The collected lithium is then released into a storage tank for further use. The clever part here is the design of the electrodes that attract many ions, but only collect and store the lithium ions, extracting the valuable element from seawater. <https://dgit.in/litdem>

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THIS MONTH IN SCIENCE:

Delve into the mechanisms of ageing and if it could be reversed. In R&D we look into how humans sense pressure and temperature. After that, head on over to 1000-words to check out some carnivorous plants.

Exploring the mechanisms of motivation

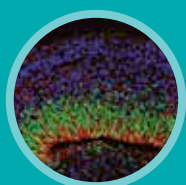
Researchers studied participants going through the "Stroop test", which involves looking at colours of words in different colours, and accurately naming the colours. The researchers then checked the accuracy of the participants under a range of variables. One measurement was the efficacy, where participants were awarded for fast, accurate answers, as against the performance not mattering



towards the prize. The prize also had a low and high amount. The study found that it was not just the amount of prize money to be won that alone determined the performance, but that efficacy was also a factor, and that

people were better motivated to perform better when they believed that their actions had a bigger role to play in the outcome. The researchers were able to isolate clusters of neurons in the brain where they could literally see the gears in the head spinning. First, the brain would evaluate the elements of the incentive, efficacy and reward, then decide how much time to allocate to the task.

<https://dgit.in/brainsci>



Brain organoids

Scientists have grown "organoids" in laboratories that mimic the structures found within real biological brains, and observed cells that behave in similar ways to the brains of those undergoing seizures, opening up the possibility of groundbreaking studies with organoids. <https://dgit.in/borga-noids>



LATR fielded

Forget the JWST, the Large Aperture Telescope Receiver, or LATR, is a cryogenically cooled astronomical instrument that will allow us to peer back into time, by studying the cosmic microwave background radiation faster and in higher resolution than ever before.

<https://dgit.in/latr>



Herd mentality

A study has found that people tend to break away from the herd when they see others hesitate before making a decision. For example, if a person sees all their friends quickly opting for a particular vaccine, they would have less hesitation in making the same decision as well.

<https://dgit.in/hm>



Purifying tablet

A hydrogel tablet has been developed that can purify water, by breaking down the cellular machinery for metabolism within bacteria. The purifying hydrogel tablets do not require an active energy source, do not create any harmful byproducts, and do not leave behind any residue.

<https://dgit.in/purhyd>